Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\det\left(\mathbb{D}_{\lambda}^{2} i.[(\nabla_{\lambda} i)^{-T} \nabla_{\lambda} g]\right) = k^{2} e^{k \log^{2} \frac{\lambda_1}{\lambda_2}} \left(\log^{2} \frac{\lambda_1}{\lambda_2}\right) \frac{1}{(\lambda_1 - \lambda_2)^{2}} \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2}\right)^{2} \geq 0.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\frac{\xi|\xi|^{-1}\varphi_t(\xi)\chi(\xi-\eta,\eta) - \eta|\eta|^{-1}\varphi_t(\eta)\chi(\eta-\xi,\xi)}{|\xi|^{3/2} - |\xi-\eta|^{3/2} - |\eta|^{3/2}} = O\big(|\eta|^{-3/2}\mathbf{1}_{[2^2,\infty]}(|\eta|/|\xi-\eta|)\mathbf{1}_{[(1+t)^{-2},0]}(|\xi|)\big),$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\mathcal{E}(d) = \sum_{i \in Q_0^{\sigma}} \frac{d_i(d_i - s_i)}{2} + \sum_{i \in Q_0^{+}} d_{\sigma(i)} d_i - \sum_{\sigma(i) \xrightarrow{\alpha} i \in Q_1^{\sigma}} \frac{d_i(d_i + \tau_{\alpha} s_i)}{2} - \sum_{i \xrightarrow{\alpha} j \in Q_1^{+}} d_{\sigma(i)} d_j.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\tilde{R}'_{11} = \left(\frac{\tilde{R}}{R}\right)^{1/2} \frac{l_p^3}{R_8 R_9} \quad , \quad \tilde{l}_p = \left(\frac{\tilde{R}}{R}\right)^{1/3} \frac{l_p^2}{(R R_8 R_9)^{1/3}} \quad , \quad \tilde{R}'_i = \left(\frac{\tilde{R}}{R}\right)^{1/2} R_i.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\left(f_{\mathcal{S}}\right)_{\lambda}(s_1, s_2) = \sup_{(y'_1, y'_2, x') \in \mathcal{S}} \left\{ f_1(y'_1) + f_2(y'_2) - \sum_{1 \leq \ell \leq 2} \lambda_{\ell} c_{\ell}\left((y_{\ell}, x_{\ell}), (y'_{\ell}, x')\right) \right\}.$$